Lab - Week 8 (Master Solutions)

Classical Game Theory (Nash Equilibrium)

1. Dominated Strategies

Tasks

3.1 Define formally what a dominated strategy is.

Answer 3.1: For Player 1, a strategy s is dominated by another strategy t is for all possible choices of actions by Player 2, the payoff for t is at least as good as (greater or equal to) the payoff of s and better for at least one choice.

3.2 Explain why a dominated strategy is an important concept.

Answer 3.2: A rational player would never play a dominated strategy

3.3 We have defined the game *Morra* in the lecture. Which strategies are dominated in *Morra*? For this, recall that the payoff matrix for (our version of *Morra*) is:

$$P = \begin{pmatrix} 0 & 2 & -3 & 0 \\ -2 & 0 & 0 & 3 \\ 3 & 0 & 0 & -4 \\ 0 & -3 & 4 & 0 \end{pmatrix};$$

Answer 3.2 : Morra does not have any dominated strategies.

3.3 (optional) For each of the bi-matrix games below, name the dominated strategies.

$$A = \begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix}; \quad B = \begin{pmatrix} 1 & 1 \\ 1 & 3 \end{pmatrix};$$ A and B are bi-matrix of the following utility matrix

$$\begin{array}{cc} & B \\ A & \begin{matrix} 2,1 & 1,1 \\ 1,1 & 1,3 \end{matrix} \end{array}$$

$$A = \begin{pmatrix} 2 & 1 & 3 & 17 \\ 27 & 3 & 1 & 1 \\ 4 & 6 & 7 & 18 \end{pmatrix}; \quad B = \begin{pmatrix} 11 & 9 & 10 & 22 \\ 0 & 1 & 1 & 0 \\ 2 & 10 & 12 & 0 \end{pmatrix}$$

$$A = \begin{pmatrix} 3 & 3 & 2 \\ 2 & 1 & 3 \end{pmatrix}; \quad B = \begin{pmatrix} 2 & 1 & 3 \\ 2 & 3 & 2 \end{pmatrix};$$

Answer 3.3:
First example: Strategy 2 for Player A, Strategy 1 for Player B
Second example: Strategy 1 for Player A (by 3), Strategy 2 for Player B (by 3)
Third example: none for Player A, Strategy 1 for Player B (by 3)

3.4 (optional) Find out what the difference is between strictly dominated and weakly dominated. Which of the dominated strategies from 3.3 are weakly dominated, which strictly?

Answer 3.4 :What we have discussed so far are weakly dominated strategies. A strictly dominated strategy delivers a strictly inferior payoff (less than) for all possible actions of the other player. Only Strategy 1 for Player A (by 3) in the second example is strongly dominated (by 3).

4. Nash Equilibrium

Nash Equilibrium in Pure Strategies

A pure Nash Equilibrium contains only one strategy of each player.

Tasks

4.1 Does the game in bi-matrix notation below have a pure Nash equilibrium? Give the equilibrium strategy profile or show why there is no pure Nash equilibrium.

$A = \left(\begin{array}{c|c} 1 & 2 \\ \hline 2 & 1 \end{array}\right);\ B = \left(\begin{array}{c|c} 4 & 2 \\ \hline 3 & 6 \end{array}\right);$

Answer 4.1: no pure Nash. For Player A: on the cells with payoff 1, Player A would unilaterally move to the other strategy. However, neither of these other outcomes is a candidate for Nash, since Player B would unilaterally switch strategy.

4.2 Does the game from 4.1 have a mixed Nash equilibrium? Give reasons.

Answer 4.2: Yes (without further analysis): Every NF game with finitely many strategies has a Nash equilibrium. To determine mixed Nash follow the procedure in the seminar to equalise the expected payoffs for each player: A-> (3/5, 2/5), B->(1/2, 1/2) **(Check the last page for full solution)**

4.3 (optional) Define a two-player zero sum game that has a pure Nash equilibrium.

Answer 4.3

For example:

In[ ]:=
```
A = (1|2 / 0|1);
TableForm[A, TableHeadings → {{"A", "B"}, {"A", "B"}}]
```

Out[ ]//TableForm=

	A	B
A	1	2
B	0	1

(A/A) is a pure Nash equilibrium.

4.2 Does the game from 4.1 have a mixed Nash equilibrium for the following utility Bi-matrix? Give the equilibrium point or give reasons why mixed Nash does not exist.

$$A = \left[\begin{array}{c|c} 1 & 2 \\ \hline 2 & 1 \end{array}\right]; \quad B = \left[\begin{array}{c|c} 4 & 2 \\ \hline 3 & 6 \end{array}\right]$$

Solution:

The above Bi-matrix can be visualized as single utility matrix with payoffs of 2-players represented as tuples,

Utility matrix is

$$\begin{array}{cc} & B \\ A & \left[\begin{array}{c|c} 1,4 & 2,2 \\ \hline 2,3 & 1,6 \end{array}\right] \end{array}$$

To find mixed Nash equilibrium, we first need to write down the expected utility for each player. Assuming there are two actions (L and R) such that the utility matrix is,

		p	$p-1$
		L	R
q	L	1, 4	2, 2
$q-1$	R	2, 3	1, 6

- The expected utility for the B (column) player is,

$$p\underbrace{[4q + 3(1-q)]}_{u_2(\mathrm{L})} + (1-p)\underbrace{[2q + 6(1-q)]}_{u_2(\mathrm{R})}$$

- The expected utility for the A (row) player is,

$$q\underbrace{[1p + 2(1-p)]}_{u_1(\mathrm{L})} + (1-q)\underbrace{[2p + 1(1-p)]}_{u_1(\mathrm{R})}$$

1. We start with player B (column). B wants to make A indifferent between L & R,

$$\begin{aligned} u_1(\mathrm{L}) &= u_1(\mathrm{R}) \\ 1p + 2(1-p) &= 2p + 1(1-p) \\ 1p + 2 - 2p &= 2p + 1 - p \\ 2 - p &= p + 1 \\ p &= \frac{1}{2} \end{aligned}$$

   Thus, player B will play (L, R) with probabilities $\left(\frac{1}{2}, \frac{1}{2}\right)$

2. Similarly, Player A wants to make B indifferent between L & R,

$$\begin{aligned} u_2(\mathrm{L}) &= u_2(\mathrm{R}) \\ 4q + 3(1-q) &= 2q + 6(1-q) \\ 4q + 3 - 3q &= 2q + 6 - 6q \\ 1q + 3 &= 6 - 4q \\ 5q &= 3 \\ q &= \frac{3}{5} \end{aligned}$$

   Thus, player A will play (L, R) with probabilities $\left(\frac{3}{5}, \frac{2}{5}\right)$